Chayanth Suthahar

942-380-8902 ✉ chayanthsuthahar@gmail.com linkedin.com/in/chayanth
Mechanical Engineering student seeking a Summer 2027 internship / co-op

Experience

Design Member & Strategy Lead 2021 – 2025
FIRST Robotics Competition Team 2056 (OP Robotics) *Stoney Creek, ON*

- Designed the drive base, gearboxes, and mounts (**10+** parts) for a **135 lb** robot, from concept to 2D drawings.
- Machined much of each robot from 2022 to 2025 on mill, lathe, and CNC; assembled arms and grippers.
- Root-caused a polycarbonate intake plate bent by impacts in 2025 practice and added a **1/2 in** round-tube ram bar to protect it; designs survived **100+** matches of robot-to-robot impact.
- As Strategy Lead for two years, was the only student on strategy, co-leading with a mentor to plan matches with **120+** teams. Team was **1st in Canada (2022–2025)** and **2025 FIRST World Champion**.
- Taught newer students design and manufacturing on a **20+** student team and opened up design resources to them.

Machine Learning Research Intern Jun. 2026 – Aug. 2026
University of Cambridge Research Institute (Sealth) *Montelaterone, Tuscany, Italy*

- Under a Cambridge professor, built a solo **Python** pipeline on AWS from Sentinel-2 satellite imagery to a random forest model, validated against ground surveys, estimating household wealth in rural South Africa.
- Extended it to nowcasting and **GDP estimates** for African countries; the institute is pitching it to governments.

Projects & Builds

"Lightning" — 2025 FRC World Championship Drivetrain *SolidWorks, CNC, Sheet Metal | 2025*

- Designed to the requirements (agile, rigid, low centre of mass): **3 in** × **1 in** × **1/8 in** aluminum tube frame and a **1/16 in** baseplate as shear plate and electronics mount, in a **32 in** × **28 in** footprint.
- Chose the **L2 ratio** on SDS MK4i modules with Kraken X60s for **15 ft/s**, balancing speed and acceleration for short cycles; a custom shifted plate with a **12-tooth** pinion for the X44 steering motors saved nearly **2 lb**.
- Stopped baseplate sag with a riveted aluminum angle near the battery; bolted the corners to back up the rivets.

RC Aircraft — RC3 Competition (Co-Founder & Design Member) *SolidWorks, FEA, 3D Printing | 2025–2026*

- Co-founded the team and helped design a **5.5 ft** wingspan tail-dragger: tapered wing on aluminum spars with pocketed ribs, 3D-printed fuselage, three carbon fibre tail booms, and aluminum landing gear with printed wheels.
- Ran **static FEA** on the wing, tail, and landing gear and iterated through several prototypes. Flew **3 min 40 s** until the battery drained to take **1st place** and **Best Handling Aircraft**.
- Helped wire the electronics for the team's Raspberry Pi Zero vision payload, which scored the full **100-point** bonus.

"VitalGroove" — Hug-Detecting Candy Dispenser (MakeUofT) *Jetson Nano, ESP32, Python | 2026*

- Designed and built the ESP32-controlled dispenser and led hardware integration: **I2C** sensors, power, and a serial link from the Jetson Nano vision pipeline that triggers a dispense. Working demo within the time limit.

"EdgeWave" — Gesture-Controlled Synthesizer (1st Place, Gibberish Hackathon) *JavaScript, MediaPipe | 2025*

- With a team of three, built a browser instrument that plays chords and drums from webcam hand tracking.

Skills

CAD & Drawings: SolidWorks (200+ part assemblies, 2D detail drawings), Onshape, Fusion 360, AutoCAD, GD&T
Analysis & Test: Static FEA (SolidWorks Simulation), prototyping, gear and motor selection, weld and material testing
Fabrication: CNC, mill, lathe, sheet metal, riveting, carbon fibre and aluminum structures, FDM 3D printing
Hardware & Code: ESP32, Arduino, Raspberry Pi, Jetson, I2C, serial, KiCad, Python, PyTorch, OpenCV, JavaScript, AWS

Leadership

Design Co-Lead Sept. 2026 – Present
TMARK (Toronto Metropolitan ARC Robotics) *Toronto, ON*

- Co-leading design of a **17 mm** launcher robot for ARC Championship Season 2 on a **60-member** team.

Co-Founder May 2026 – Present
Sandbox *Toronto, ON*

- Run hackathons with a **15-person** team; planned an official Toronto Tech Week hackathon (**40** people) in one week.
- Partnering with Filipinos Forward on a kid-judged charity hardware hackathon funding tech in the Philippines.

Education

Toronto Metropolitan University Toronto, ON
Bachelor of Engineering (BEng), Mechanical Engineering *Sept. 2025 – Expected Apr. 2030*